# Multimodal integration reveals complementary imaging and genetic predictors of coronary artery disease risk

Devansh Pandey[1]*, Liaoyi Xu[1], Eucharist Kun[1], Chenfei Li[1], Joyce Y. Wang[1], Alaa Melek[2], Julie C. DiCarlo[3,5], Edward Castillo[2], Jagat Narula[4], Charles A. Taylor[3,5], Vagheesh M. Narasimhan[1,3,6]

[1]Integrative Biology · [2]Biomedical Engineering · [3]Oden Institute for Computational Engineering and Sciences · [5]Internal Medicine, Dell Medical School · [6]Statistics and Data Sciences — The University of Texas at Austin · [4]UT Health Science Center, Houston *Presenter: Devansh Pandey · [email placeholder]

github.com/Devanshpandey/Multimodal-integration-for-coronary-artery-disease-risk-prediction

**Cardiac MRI embeddings and polygenic risk capture independent CAD risk — combining them identifies a subgroup with ≈23× the cohort event rate.**

**Colour key** 🟧 imaging (cardiac MRI) 🟧 extra-cardiac imaging 🟦 genetics (PRS) ⬜ clinical / guideline scores ⬛ combined model

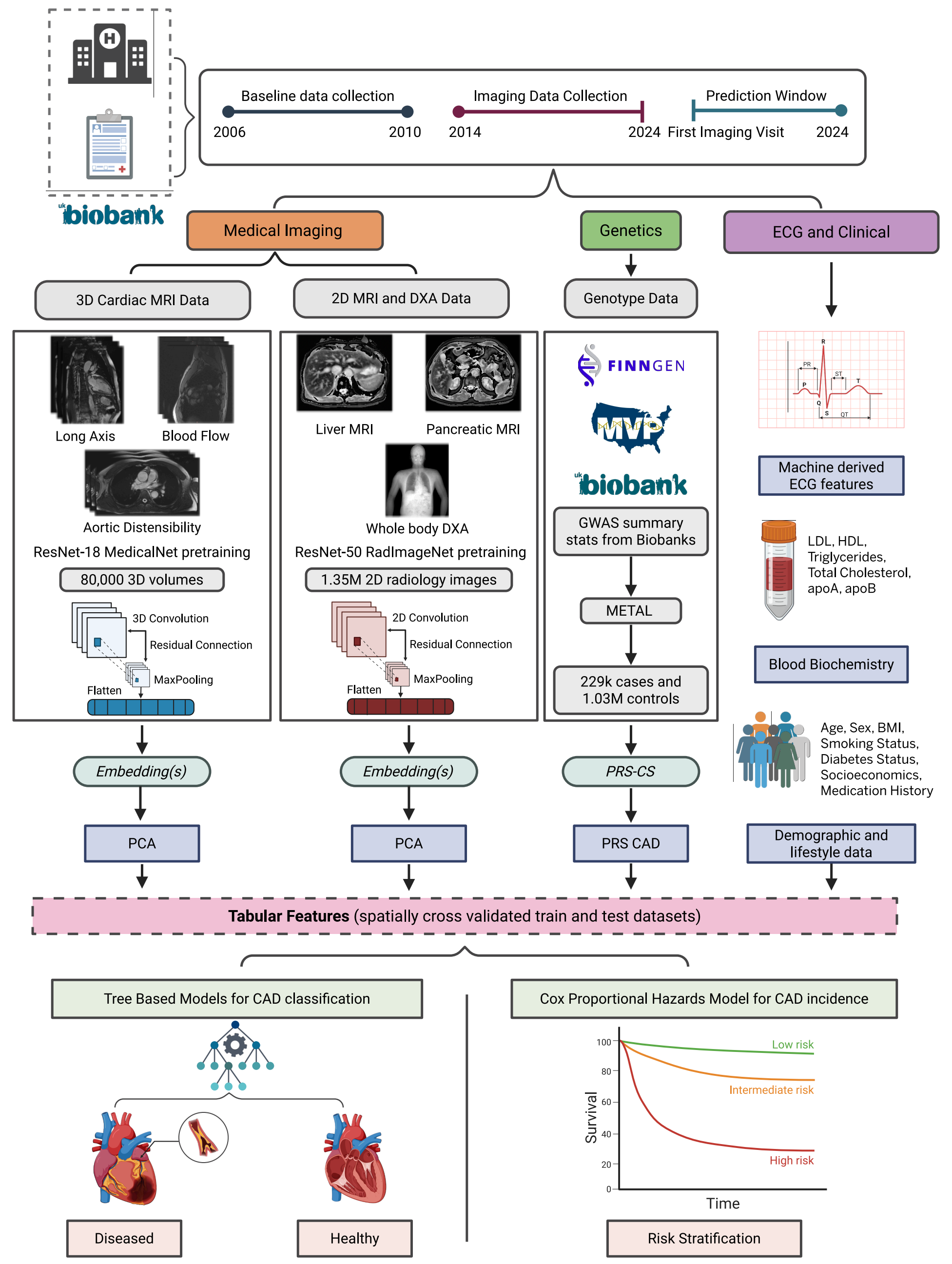
## WHY & HOW

### UK Biobank imaging cohort, three data domains, one model

### Background & questions

- Leading cause of death; ~50% of CAD first presents as a heart attack.
- Clinical scores omit genetics and imaging; how much imaging and genetic risk overlap was unknown.

1. Which imaging modalities add to clinical, metabolic, ECG and genetic factors?
2. Overlap between imaging-derived and polygenic risk
3. Sequential integration: discrimination, calibration, stratification



**42,351**
UK Biobank imaging participants

**677**
incident CAD (1.6%)

**4.6 yr**
median follow-up

**211,755**
cardiac cine MRI series

**127,053**
2D images (liver, pancreas, DXA)

**10,597**
held-out test set (178 cases)

*Cardiac MRI does not image coronary atherosclerosis; its signal reflects remodeling, vascular stiffness and cardiometabolic burden.*

**Outcome = clinically recognized CAD** (MI, angina, revascularization); prevalent cases excluded. Fig. 1 of the manuscript: three data domains → tabular features → classification and Cox survival models.

### Methods in brief

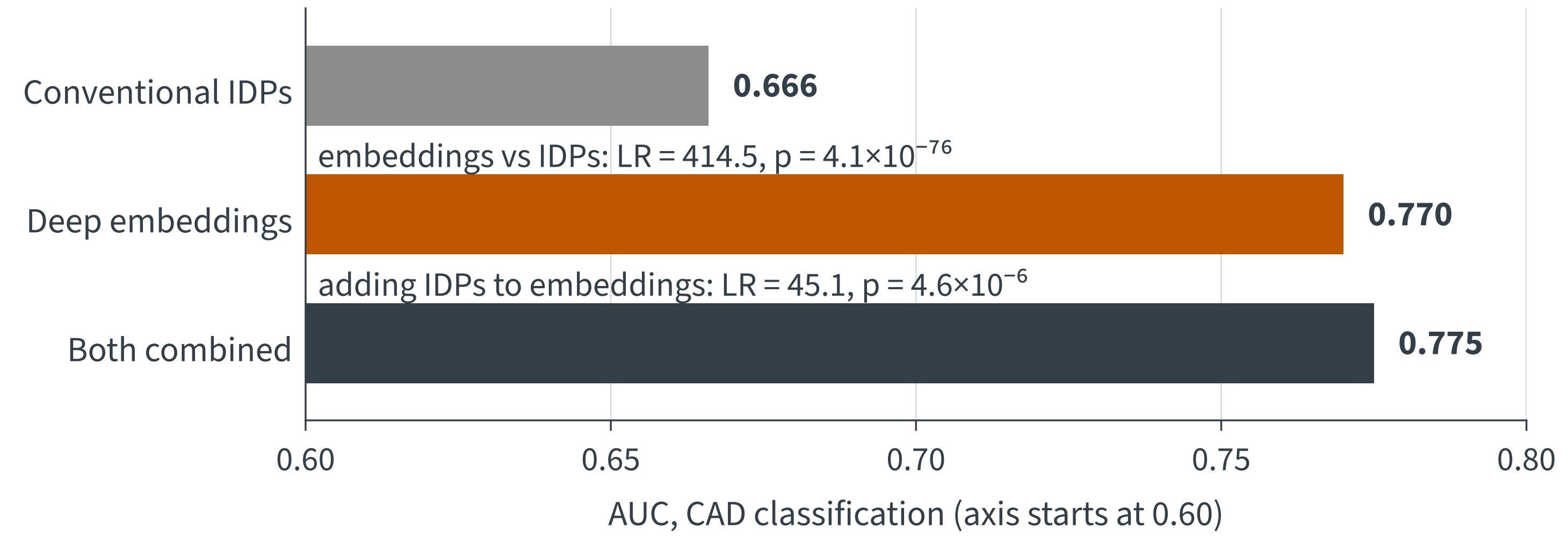
- Encoders fine-tuned on balanced case–control training data only.
- Embeddings → PCA, 10 PCs per stream (80 total).
- PRS-CS, European meta-GWAS: OR 1.87 per SD, AUC 0.766 (with age, sex, 20 PCs).
- XGBoost + nested Cox; imaging risk score = cross-fitted ridge Cox.

## IMAGING REPRESENTATIONS

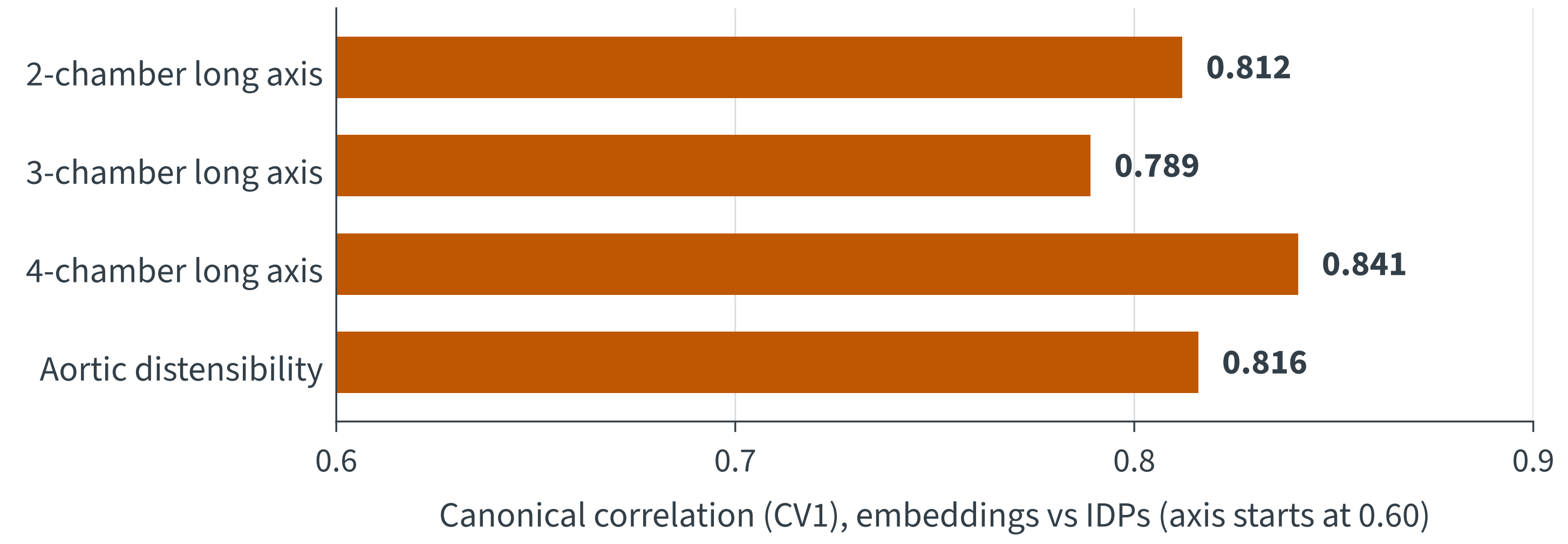
### Deep embeddings beat handcrafted cardiac phenotypes

**0.770 vs 0.666**

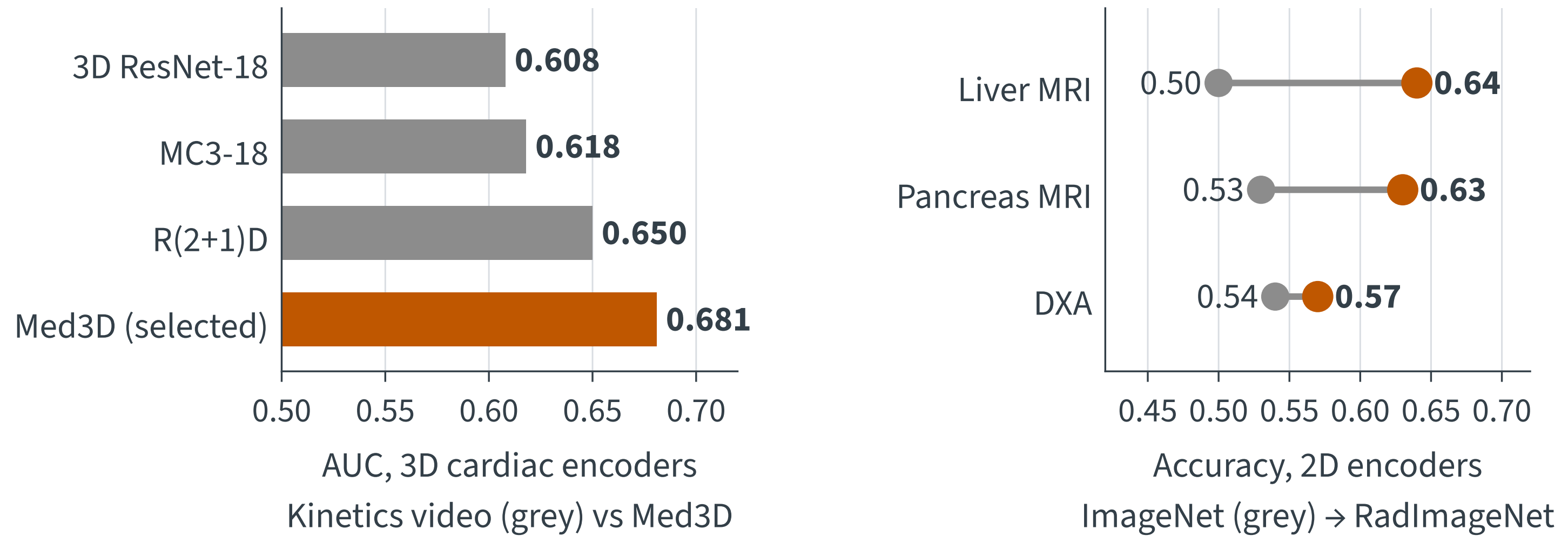
AUC, deep embeddings vs conventional imaging-derived phenotypes (IDPs); both combined 0.775



**Embeddings outperform IDPs; adding IDPs barely helps** (logistic models; performance plateaus at ~5 PCs per modality; Fig. 2a,b).



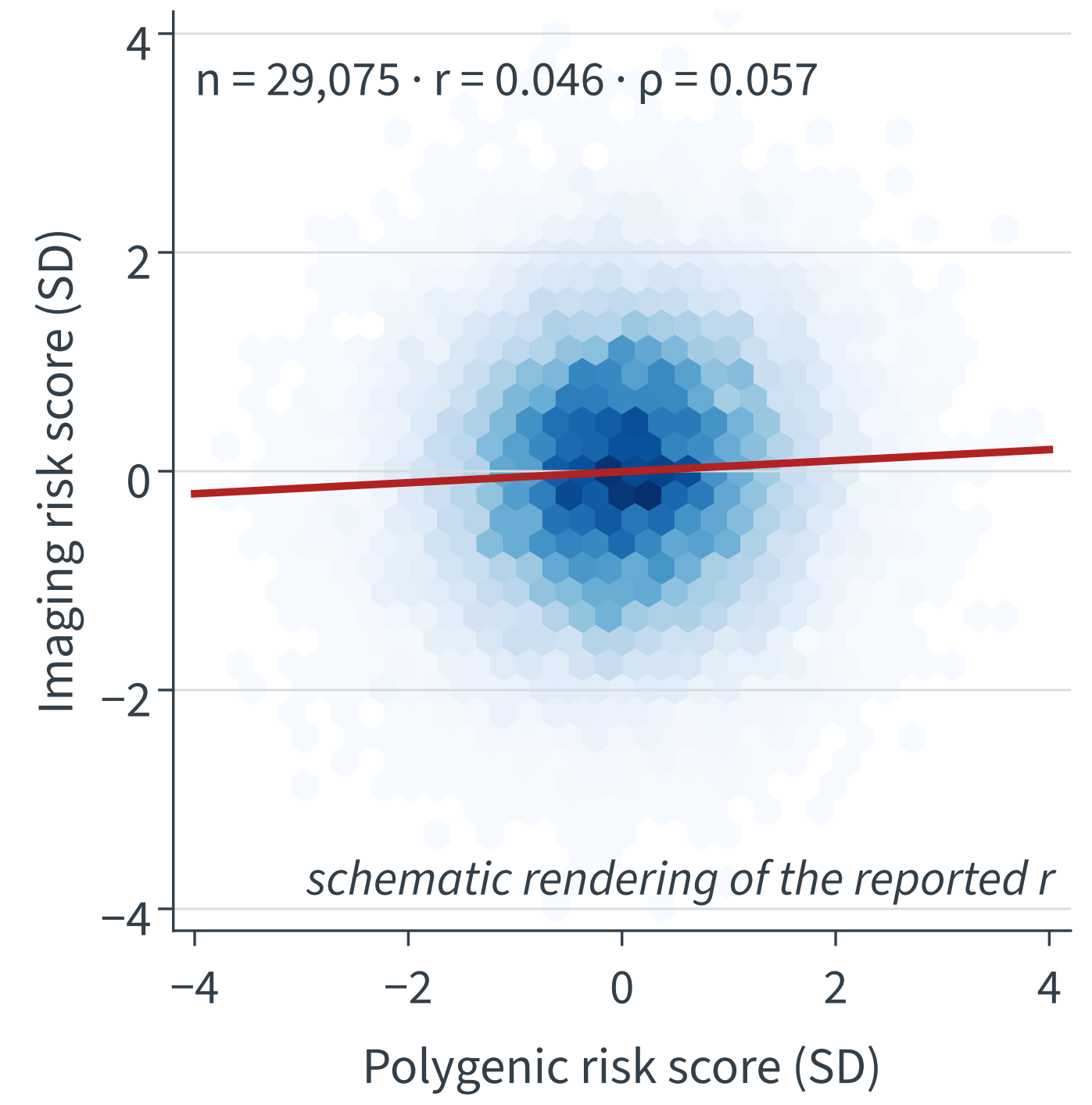
**Embeddings encode physiology:** CV1 loads on stroke volume, ejection fraction, myocardial mass and wall thickness (Fig. 2c).



**Medical pretraining wins:** Med3D for the cardiac 3D encoders, RadImageNet for liver, pancreas and DXA (axes from 0.50 / 0.45; Supp. Fig. 3b, 4b).

## IMAGING VS GENETICS

### Imaging and genetic risk are nearly independent



**r ≈ 0.05**

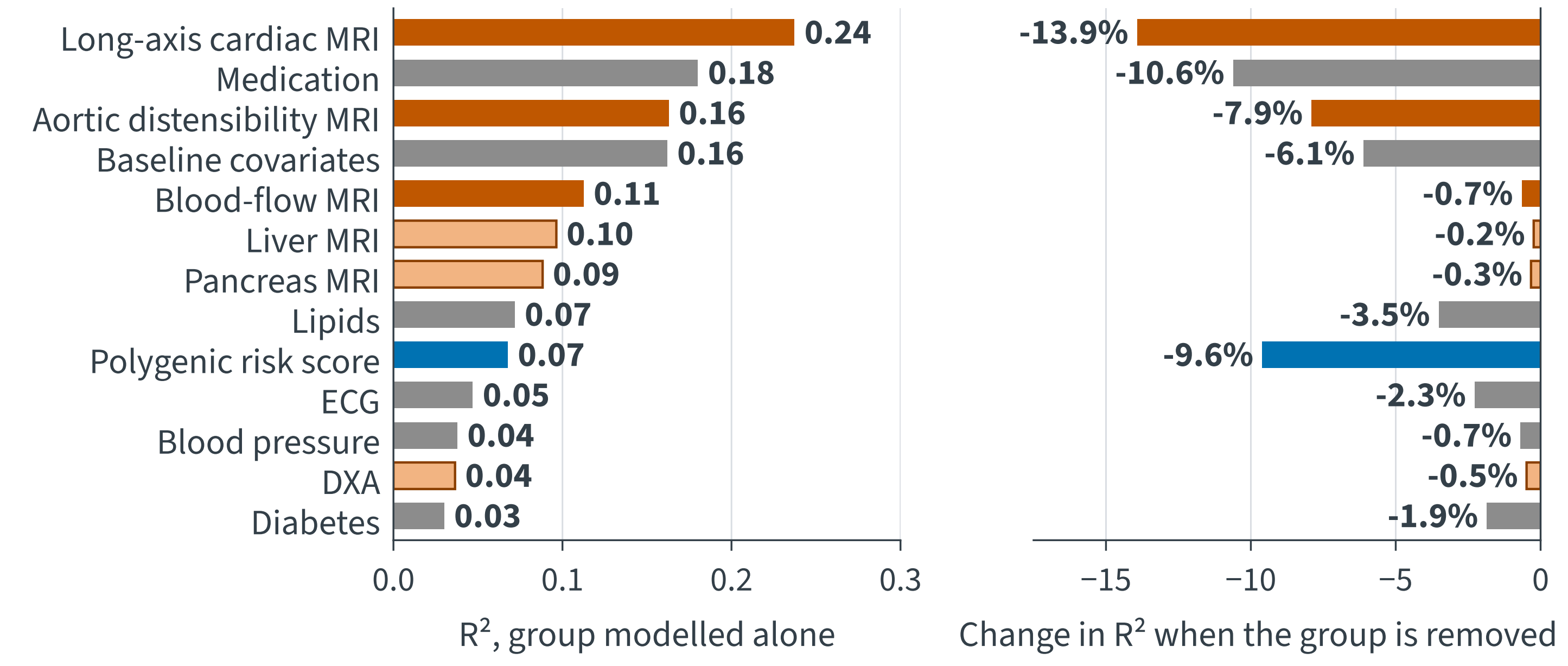
Pearson r, imaging vs polygenic risk score (ρ = 0.057); $R^2 < 1\%$ shared — the two overlap very little (n = 29,075; Fig. 3c–e).

**Partial $R^2$ of imaging risk explained by**

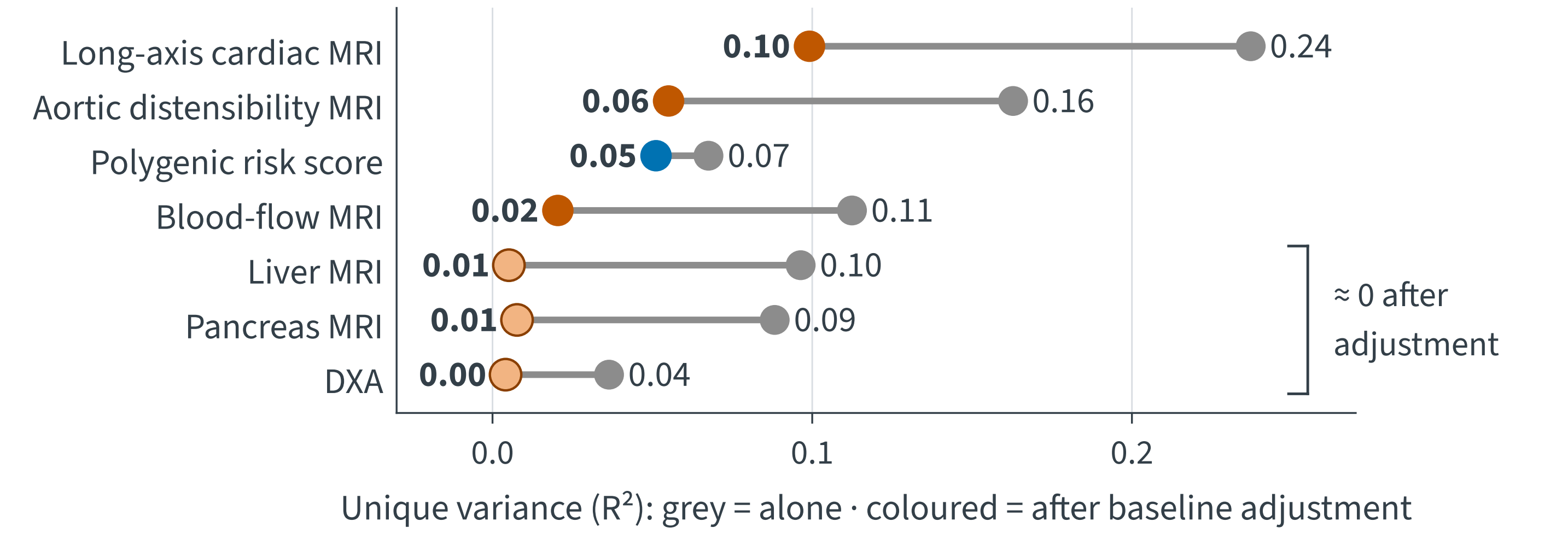
**0.15%** genetics (PRS)

**38.4%** demographics

**14.4%** clinical factors



**Cardiac imaging, medication and PRS are each non-redundant.** Left: $R^2$ of each group alone; right: $R^2$ drop when the group is removed from the full model (Fig. 3a,b).



**Only long-axis, aortic distensibility MRI and PRS retain unique variance** after clinical adjustment (Fig. 3f).

## INTEGRATION & STRATIFICATION

### Sequential integration finds very-high-risk people

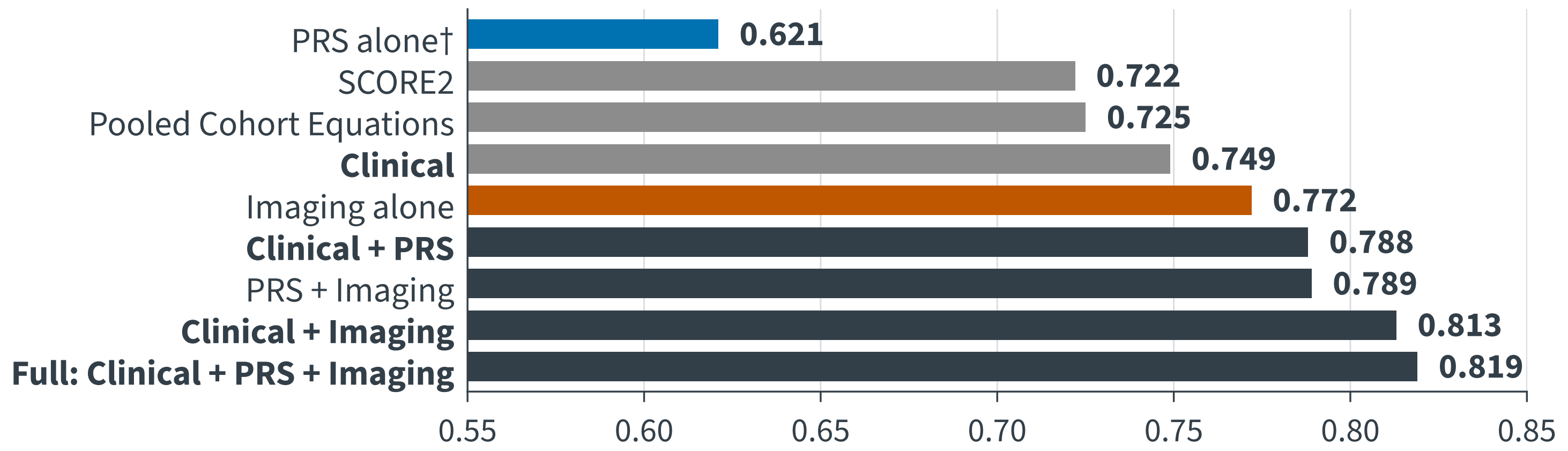
**36.8%**
5-yr CAD incidence, top quintile of both imaging and genetic risk within Tier 3; cohort 1.6%

**0.634**
continuous NRI vs clinical (95% CI 0.534–0.734)

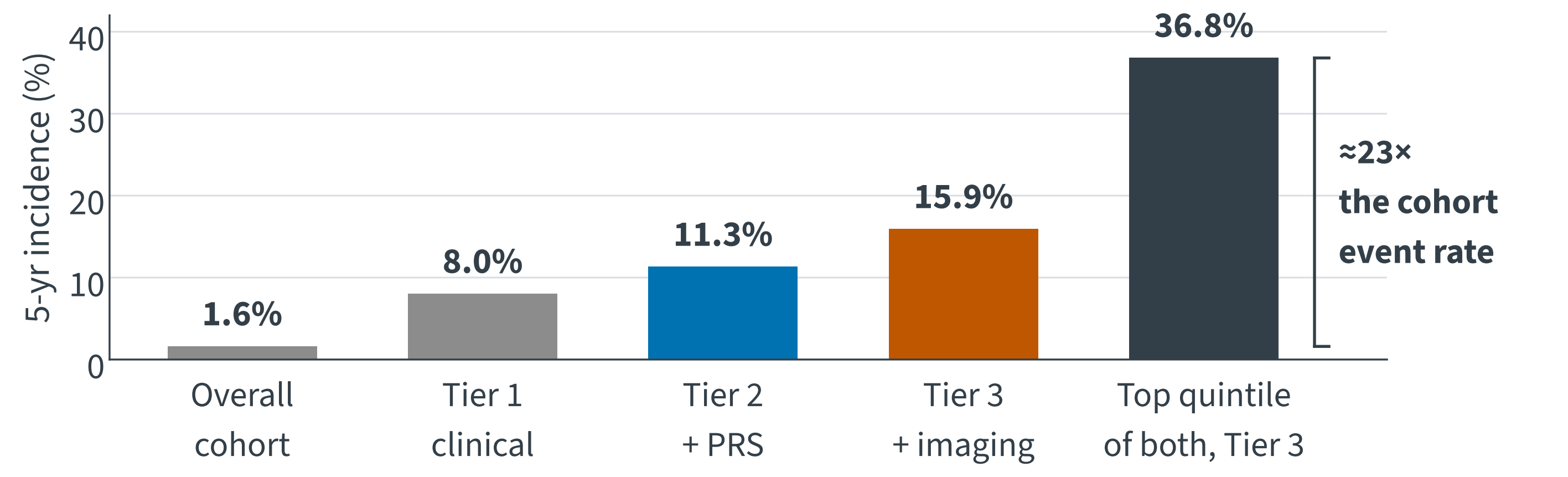
**9.5%**
prediction accuracy (IPA), full model; clinical 5.5%

**86.8 yr**
median heart age, Tier 3; cohort 55.3 yr

**0.99–1.08**
calibration slopes, nested models



**Each layer adds discrimination; imaging is among the strongest independent predictors (HR ≈ 2.1 per SD).** Held-out C-index; axis from 0.55; bold = nested. HR per SD, full model: imaging ≈ 2.1, PRS ≈ 1.7, demographics ≈ 1.4, lipids ≈ 1.1. †PRS alone = score without age/sex; AUC 0.766 (column 1) includes age, sex, 20 PCs.



**Sequential tiers enrich risk far beyond clinical stratification** (5-yr incidence, held-out set). Leave-one-centre-out: AUC 0.785–0.822, C-index 0.73–0.76; discrimination is stable across the four UK Biobank imaging centres (site-level robustness, not external validation).

### Conclusions

- Imaging and polygenic risk are nearly uncorrelated: complementary axes.
- Long-axis cine and aortic distensibility MRI carry the incremental signal; blood-flow, liver, pancreas and DXA add little once clinical variables are included.
- Clinical → PRS → imaging integration isolates a 36.8% 5-yr-risk subgroup.
- A framework for complementary risk — not yet a clinical screening tool.

### Limitations

- No CAC/CTA in UK Biobank: no calcium-score benchmark.
- European-ancestry, complete-case, healthier-than-general cohort; covariates measured at imaging, so part of imaging's gain may reflect temporal proximity.
- Not externally validated; observational — prospective trials needed.